

1 Pre-lab design

1.1 Question D1

The following model describes the changes in electrical resistance as a function of strain:

$$\epsilon = \frac{dR}{RG'} \quad (1)$$

Where:

- ϵ represents strain factor
- R is the nominal resistance
- dR is the change in resistance
- G is the gauge factor.

Calculate the change in resistance dR if $R = 120\Omega$, $G = 2$ and $\epsilon = 0.01$:

$$\begin{aligned} dR &= \epsilon RG' \\ dR &= 0.01 \cdot 120\Omega \cdot 2 \\ dR &= 2.4\Omega \end{aligned} \quad (2)$$

1.2 Question D2

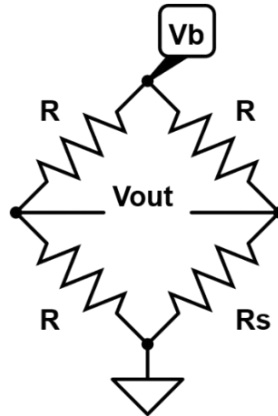


Figure 1: Wheatstone bridge.

Let the strain gauge resistance be represented by Rs :

$$Rs = R + dR \quad (3)$$

Let the other three resistors have the same nominal resistance R_s as the strain gauge. The expression for V_{out} becomes:

$$\begin{aligned}
 V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{Rs}{Rs+R} \\
 V_{out} &= V_b \left(\frac{1}{2} - \frac{Rs}{Rs+R} \right) \\
 V_{out} &= V_b \left(\frac{1}{2} - \frac{R+dR}{2R+dR} \right)
 \end{aligned} \tag{4}$$

1.3 Question D3

Consider $V_b = 5V$, $R = 120\Omega$, calculate V_{out} for a 1% strain:

$$\begin{aligned}
 V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{Rs}{Rs+R} \\
 V_{out} &= 5V \cdot \frac{120\Omega}{240\Omega} - 5V \cdot \frac{120\Omega + 2.4\Omega}{120\Omega + 120\Omega + 2.4\Omega} \\
 V_{out} &= 5V \left(\frac{1}{2} - \frac{122.4\Omega}{242.4\Omega} \right) \\
 V_{out} &= -0.02475V
 \end{aligned} \tag{5}$$

Re-calculate now where each bridge resistor features $R = 114\Omega$:

$$\begin{aligned}
 V_{out} &= V_b \cdot \frac{R}{R+R} - V_b \cdot \frac{Rs}{Rs+R} \\
 V_{out} &= 5V \cdot \frac{114\Omega}{228\Omega} - 5V \cdot \frac{120\Omega + 2.4\Omega}{120\Omega + 114\Omega + 2.4\Omega} \\
 V_{out} &= 5V \left(\frac{1}{2} - \frac{122.4\Omega}{236.4\Omega} \right) \\
 V_{out} &= -0.08883V
 \end{aligned} \tag{6}$$

1.4 Question D4